Exercise JA

Q1. Consider the system of equations

$x - 2y + 3z = -1, - x + y - 2z = k, x - 3y + 4z = 1$

Statement-1 : The system of equations has no solution for k ¹ 3.

and

Statement-2 : The determinant $\begin{bmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{bmatrix} \neq 0$, for $k \neq 3$.

(A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1

(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Ans. 5ee893a39029e46015256915

Sol. $x - 2y + 3z = -1, - x + y - 2z = k$ & $x - 3y + 4z = 1$

$$\Delta = \begin{vmatrix} 1 & 3 & -1 \\ -1 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix} = 0 \ \& \ \begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} = 3 - k$$

Hence if k = 3 then system will have infinite solutions and k 3 then system will have no solution. So S(I) & S (II) both are true & (II) is correct explaination for (I)

Q2. The number of all possinle values of θ, where $0 < \theta < \pi$, for which the system of equations

$(y + z) \cos 3\theta = (xyz) \sin 3\theta$

$x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$

$(xyz) \sin 3\theta = (y + 2z) \cos 3\theta + y \sin 3\theta$

have a solution $(x_0, y_0\, z_0)$ with $y_0\, z_0 \neq 0$, is

Ans. 3

Sol. $(y + z) \cos 3\theta = (xyz) \sin 3\theta$ ....(i)

$\sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$ ....(ii)

$(xyz) \sin 3\theta = (y + 2z) \cos 3\theta + y \sin 3\theta$ ....(iii)

from (i) & (iii)

$(y + z) \cos 3\theta = (y + 2z) \cos 3\theta + y \sin 3\theta$

$\Rightarrow$ z cos 3θ + y sin $3\theta = 0$ ....(iv)

from equation (ii)

2z cos 3θ + 2y sin 3θ = xyz sin 3θ ....(v)

from equation (iv) & (v)

xyz sin $3\theta = 0$

$\Rightarrow$ x sin $3\theta = 0$ as yz $\neq 0$

Possible cases are either x = 0 or sin $3\theta = 0$

Case (1) : if x = 0

$\Rightarrow$ y + z = 0 $\Rightarrow$ y = – z

from equation (iv) cos 3θ = sin 3θ

$\Rightarrow 3\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}$

Case (2) : if sin $3\theta = 0$

$\Rightarrow \theta = \frac{\pi}{3}, \frac{2\pi}{3}$

But these values does not satisfy given equations.
Hence, total number of possible values of q are 3.

Q3. Which of the following values of α satisy the equation $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (1+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix}$ = –648 α ?

(A) –4 (B) 9 (C) –9 (D) 4

Ans. B, C

Sol. $R_3 \to R_3 - R_2, R_2 \to R_2 - R_1$

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 5+2\alpha & 5+4\alpha & 5+4\alpha \end{vmatrix} = 648\alpha$$

$R_3 \to R_3 - R_2$

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 2 & 2 & 2 \end{vmatrix} = 648\alpha$$

$C_3 \to C_3 - C_2$, $C_2\ C_2 - C_1$

$$\begin{vmatrix} (1+\alpha)^2 & \alpha(2+3\alpha) & \alpha(2+5\alpha)^2 \\ 3+2\alpha & 2\alpha & 2\alpha \\ 2 & 0 & 0 \end{vmatrix} = -648\alpha$$

$\Rightarrow 2\alpha^2(2 + 3\alpha) - 2\alpha^2\,(2 + 5\alpha) = -324\alpha$

$\Rightarrow -4\alpha^3 = 324\alpha \qquad \Rightarrow \alpha = 0, \pm\ 9$

Q4. The total number of distinct x $\in$ R for which $\begin{bmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{bmatrix} = 10$ is.

Ans. 2

Sol. $\begin{vmatrix} x & x^2 & 1 \\ 2x & 4x^2 & 1 \\ 3x & 9x^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 2x & 4x^2 & 1 \\ 8x^3 & 9x^2 & 27x^3 \end{vmatrix} = 10$

Using $\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} = (a - b)\,(b - c)\,(c - a)$

Again $\begin{vmatrix} x & x^2 & 1 \\ 2x & 4x^2 & 1 \\ 3x & 9x^2 & 1 \end{vmatrix} + (x)\,(2x)\,(3x) \begin{vmatrix} 1 & x & x^2 \\ 1 & 2x & 4x^2 \\ 1 & 3x & 9x^2 \end{vmatrix} = 10$

$\Rightarrow (x - 2x)\,(2x - 3x)\,(3x - x) + 6x^3\,[(x - 2x)\,(2x - 3x)\,(3x - x)] = 10$

$\Rightarrow 2x^3\,(1 + 6x^3) = 10$ (solve by putting $x^3 = t$)

$\Rightarrow x = -1, \left(\frac{5}{6}\right)^{\frac{1}{3}}$

Q5. Let a, λ, $\mu \in$. Consider the system of linear equations

$ax + 2y = \lambda$

$3x - 2y = \mu$

Which of the following statement(s) is(are) correct ?

(A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ

(B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

Ans. D, C, B

Sol. System has unique solution for $\frac{a}{3} \neq \frac{2}{-2}$

system has infinity many solution for $\frac{a}{3} = \frac{2}{-2} = \frac{\lambda}{\mu}$ and no solution for $\frac{a}{3} = \frac{2}{-2} \neq \frac{\lambda}{\mu}$

Q6. Let p, q, r be nonzero real numbers that are, respectively, the 10^{th}, 100^{th} and 1000^{th} terms of a harmonic progression. Consider the system of linear equations

$$x + y + z = 1$$
$$10x + 100y + 1000z = 0$$
$$qr\, x + pr\, y + pq\, z = 0$$

List-I	**List-II**
(I) If $\frac{q}{r} = 10$, then the system of linear equations has	(P) $x = 0,\ y = \frac{10}{9}, z = -\frac{1}{9}$ as solution
(II) If $\frac{p}{q} \neq 100$, then the system of linear equations has	(Q) $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ as solution
(III) If $\frac{p}{q} \neq 10$, then the system of linear equations has	(R) infinitely many solutions
(IV) If $\frac{p}{q} = 10$, then the system of linear equations has	(S) no solution
	(T) at least one solution

The correct option is :

(A) (I) → (T); (II) → (R); (III) → (S); (IV) → (T)

(B) (I) → (Q); (II) → (S); (III) → (S); (IV) → (R)

(C) (I) → (Q); (II) → (R); (III) → (P); (IV) → (R)

(D) (I) → (T); (II) → (S); (III) → (P); (IV) → (T)

Ans. 63133c2ef69e6e3404af1087

Sol. $x + y + z = 1$ ________(1)

$10x + 100y + 1000z = 0$ ________(2)

$qrx + pry + pqz = 0$ ________(3)

Equation (3) can be re-written as

$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 0$ $(\because$ p, q, r $\neq 0)$

Let $p = \frac{1}{a+9d}, q = \frac{1}{a+99d}, r = \frac{1}{a+999d}$

Now, equation (3) is

(a + 9d)x + (a + 99d)y + (a + 999d)z = 0

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 100 & 1000 \\ a+9d & a+99d & a+999d \end{vmatrix} = 0$$

$$\Delta_x = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 100 & 1000 \\ 0 & a+99d & a+999d \end{vmatrix} = 900(d-a)$$

$$\Delta_y = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 0 & 1000 \\ a+9d & 0 & a+999d \end{vmatrix} = 990(a-d)$$

Option I: If $\frac{q}{r} = 10 \Rightarrow a = d$

$\Delta = \Delta_x = \Delta_y = \Delta_z = 0$

And eq. (1) and eq. (2) represents non-parallel planes and eq. (2) and eq. (3) represents same plane $\Rightarrow$ Infinitely many solutions.

I $\to$ P, Q, R, T

Option II : $\frac{p}{r} \neq 100 \Rightarrow a \neq d$

$\Delta = 0, \Delta_x, \Delta_y, \Delta_z \neq 0$

No solution

II $\to$ S

Option III : $\frac{p}{q} \neq 10 \Rightarrow a \neq d$

No solution

III $\to$ S

Option IV : If $\frac{p}{q} = 10 \Rightarrow a = d$

Infinitely many solution

IV $\to$ P, Q, R, T

Q7. Let α, β and γ be real numbers. Consider the following system of linear equations

x + 2y + z = 7

x + αz = 11

2x – 3y + βz = γ

Match each entry in List-I to the correct entries in.

List-I	List-II
(P) If $\beta = \frac{1}{2}(7\alpha - 3)$ and γ = 28, then the system has	(1) a unique solution
(Q) If $\beta = \frac{1}{2}(7\alpha - 3)$ and γ ≠ 28, then the system has	(2) no solution
(R) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where α = 1 and γ ≠ 28, then the system has	(3) infinitely many solutions
(S) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where α = 1 and γ = 28, then the system has	(4) x = 11, y = –2 and z = 0 as a solution

	(5) $x = -15$, $y = 4$ and $z = 0$ as a solution

The correct option is :

(A) (P) → (3); (Q) → (2); (R) → (1); (S) → (4)

(B) (P) → (3); (Q) → (2); (R) → (5); (S) → (4)

(C) (P) → (2); (Q) → (1); (R) → (4); (S) → (5)

(D) (P) → (2); (Q) → (1); (R) → (1); (S) → (3)

Ans. 650a9db9f86329931f85e28a

Sol. $$D = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & \alpha \\ 2 & -3 & \beta \end{vmatrix}$$

$= 1\,[3\alpha] - 2\,[\beta - 2\alpha] + [-3]$

$= 3\alpha - 2\beta + 4\alpha - 3$

$= 7\alpha - 2\beta - 3$

If $D = 0$

$7\alpha - 2\beta - 3 = 0$

$$\beta = \frac{7\alpha - 3}{2}$$

$$D_x = \begin{vmatrix} 7 & 2 & 1 \\ 11 & 0 & \alpha \\ \gamma & -3 & \beta \end{vmatrix}$$

if $\gamma = 28$ & $\beta = \frac{7\alpha - 3}{2}$

then $D_x = \begin{vmatrix} 7 & 2 & 1 \\ 11 & 0 & \alpha \\ 28 & -3 & \frac{7\alpha-3}{2} \end{vmatrix}$

$$7\,[3\alpha] - 2\left[11\frac{(7\alpha - 3)}{2} - 28\alpha\right] + [-33]$$

$21\alpha - [11\,(7\alpha - 3) - 56\alpha] - 33$

$21\alpha - (77\alpha - 33 - 56\alpha) - 33$

$= 0$

$$D_y = \begin{vmatrix} 1 & 7 & 1 \\ 1 & 11 & \alpha \\ 2 & \gamma & \beta \end{vmatrix}$$

Again if $\gamma = 28$ & $\beta = \frac{7\alpha - 3}{2}$

$$D_y = \begin{vmatrix} 1 & 7 & 1 \\ 1 & 11 & \alpha \\ 2 & 28 & \frac{7\alpha-3}{2} \end{vmatrix}$$

$R_1 \to R_1 - R_2$

$$= \begin{vmatrix} 0 & -4 & 1-\alpha \\ 1 & 11 & \alpha \\ 2 & 28 & \frac{7\alpha-3}{2} \end{vmatrix}$$

$$4\left[\frac{7\alpha-3}{2} - 2\alpha\right] + (1-\alpha)(28-22)$$

$2(7\alpha - 3) - 8\alpha + 6(1 - \alpha)$

$14\alpha - 6 - 8\alpha + 6 - 6\alpha$

$= 0$

$$D_z = \begin{vmatrix} 1 & 2 & 7 \\ 1 & 0 & 11 \\ 2 & -3 & \gamma \end{vmatrix}$$

$1(33) - 2(\gamma - 22) + 7(-3)$

$33 + 44 - 21 - 2\gamma$

$\Rightarrow 56 - 2\gamma$

If $\gamma = 28$

$D_z = 0$

(P) If $\beta = \frac{1}{2}(7\alpha - 3)$ & $\alpha = 28$

then D_x, D_y, D_z are all zero

$\therefore$ Infinite many solution

$P \to (3)$

(Q) If $\beta = \frac{1}{2}(7\alpha - 3)$ & $g \neq 28$

It can be easily observed that $D_z \neq 0$

$\therefore$ $D = 0$ & $D_z \neq 0$

$\Rightarrow$ No solution

$Q \to 2$

(R) If $\beta \neq \frac{1}{2}(7\alpha - 3)$, i.e. $D \neq 0$

It will be the case of unique solution but here additional info of $\alpha = 1$ & $\gamma \neq 28$ is provided

(S) If $\alpha = 1$ then

$x + 2y + z = 7$

$x + z = 11$

$2y = 7 - 11$

$y = -2$ will always be the solution.

But $\gamma \neq 28$ then $z \neq 0$

So, $R \to (1)$

Again

If $\alpha = 1$; $\gamma = 28$

then $y = -2$; $z = 0$

& $z + 2(-2) = 7$

$x = 7 + 4$

$x = 11$

So we get the solution S $\rightarrow$ (4)